

Regression and Residuals

Project: Responsible Plant Classifier

1. Regression and Residuals

Teach the learning cycle

Primary target: Distinguish numeric prediction from classification

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using evidence

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The training-majority rule always predicts clear. Matrix $\begin{bmatrix} 16 & 0 \\ 4 & 0 \end{bmatrix}$ gives $16/20=80$ percent, $FP=0, FN=4$. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Connect training-only scaling and neighbour prediction in one pipeline

Reference example, not the only acceptable wording: fit learns training minima and ranges and stores scaled training rows with IDs and labels. predict takes a raw feature pair and the fitted state. Evaluation compares a prediction with its reference label afterward. Validation references belong only in evaluation, not in fitting or the neighbour pool.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Actual leaf length is 7 cm and prediction is 10 cm. Using $\text{residual} = \text{actual} - \text{predicted}$, calculate residual and absolute error. Interpret the sign.

Concept to explain

Regression predicts a number; a residual records signed error and reveals direction as well as size.

Term · Meaning

Term: regression

Meaning: prediction of a numerical target

Term: prediction

Meaning: model-estimated value

Term: residual

Meaning: actual value minus predicted value

Term: residual plot

Meaning: errors displayed against predictions or inputs

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on regression and residuals. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

A category; regression predicts a numeric target instead.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain regression and residuals using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Plot predictions and calculate positive and negative residuals.

MODEL IDEA Regression predicts a number; a residual records signed error and reveals direction as well as size.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual plant height is 30 cm; prediction is 27 cm. Residual = actual – predicted = $30 - 27 = +3$ cm.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Actual plant height is 30 cm; prediction is 27 cm.

STEP 2 Residual = actual – predicted = $30 - 27 = +3$ cm.

STEP 3 Another case has actual 20 and predicted 24, residual –4 cm.

STEP 4 Plot residuals; a pattern above or below zero suggests systematic model error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Actual height=12 cm, prediction=9 cm. Residual means actual minus predicted.

Calculate and interpret.

Exact explanation to model

+3 cm means underprediction by 3 cm. State this sign convention consistently.

Explain the added worked example

A school garden example predicts plant height in cm from recorded conditions. The target is numeric, so a prediction can be too high or too low by different amounts. Define residual as observed minus predicted. Keep units and the sign convention visible; some software uses the opposite error convention.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines

regression and residuals.

Observed heights [4, 7, 9] cm and predictions [5, 6, 9] cm give residuals [-1, 1, 0] cm. The first prediction is too high, the second too low and the third exact. Mean signed residual is zero, yet two predictions are wrong. Opposite errors can cancel.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: Regression

Meaning: Prediction of a numeric target

Term: Residual

Meaning: Observed target minus predicted target in this course

Term: Underprediction

Meaning: A prediction below the observed value

Term: Overprediction

Meaning: A prediction above the observed value

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Regression and residuals at a glance

Part · Evidence or action · Interpretation

Part: Observed 4 predicted 5

Evidence or action: Residual minus 1

Interpretation: Overprediction

Part: Observed 7 predicted 6

Evidence or action: Residual plus 1

Interpretation: Underprediction

Part: Observed 9 predicted 9

Evidence or action: Residual zero

Interpretation: Exact on this record

To inspect a residual plot, place a feature or prediction on the horizontal axis and residual on the vertical axis. A trend can suggest a missed relationship. It does not by itself identify the cause or validate a new model. Use development data for analysis and preserve an independent final test.

Why does mean residual zero not imply perfect predictions? Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Regression and Residuals

Use residual signs to say whether each plant-height estimate is too high or low.

The facts we will use

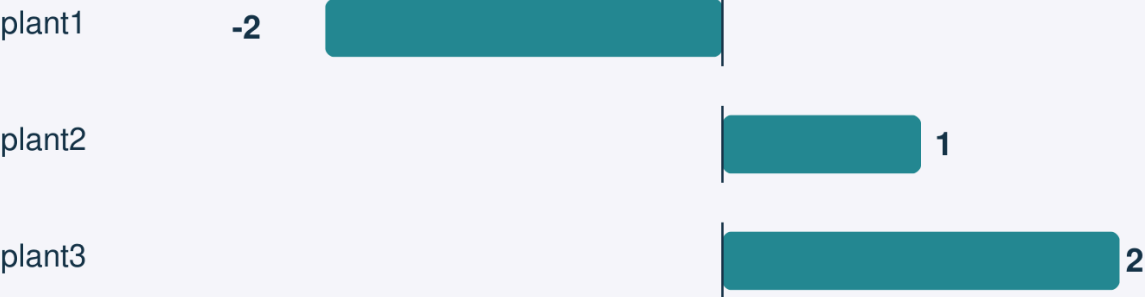
Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/19

Observed minus predicted height

cm



Bar length shows the supplied value; the marked baseline is zero.

Regression and Residuals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Subtract matched pairs

Residuals are $5-7=-2$, $8-7=1$, $12-10=2$ cm.



2. Read directions

First prediction is too high by 2; second too low by 1; third too low by 2.



3. Avoid cancellation

Mean residual is $1/3$ cm; MAE is $5/3$ cm, so small signed mean does not imply small errors.

CHANGE ONE THING

Raise all predictions by 1 cm.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Subtract matched pairs

What happens and why: Residuals are $5-7=-2$, $8-7=1$, $12-10=2$ cm.

Step: 2. Read directions

What happens and why: First prediction is too high by 2; second too low by 1; third too low by 2.

Step: 3. Avoid cancellation

What happens and why: Mean residual is $1/3$ cm; MAE is $5/3$ cm, so small signed mean does not imply small errors.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Mean residual is $1/3$ cm; MAE is $5/3$ cm, so small signed mean does not imply small errors. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Always declare the residual sign convention; some software uses predicted minus observed.

Change one thing in the pictured case

Change: Raise all predictions by 1 cm.

Prediction: Residuals become $[-3, 0, 1]$; mean $= -2/3$ cm and MAE $= 4/3$ cm.

Reason: Each residual decreases by 1 under the observed-minus-predicted convention.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Actual height $= 12$ cm, prediction $= 9$ cm. Residual means actual minus predicted.

Write the question to explain: Calculate and interpret.

Reveal after predictions: $+3$ cm means underprediction by 3 cm. State this sign convention consistently.

Keep for an independent attempt: Define residual as actual minus predicted. Actual height is 14 cm and prediction is 17 cm. Calculate residual and interpret its sign.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Negative and positive residuals cancel even when individual errors are nonzero. Exit response: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What does a class label predict? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify regression in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: calculate residuals; explain cancellation. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between regression and residual. Invite a counterexample to an

overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Interpret residuals -2, 0, 3 ordered by growth stage.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Distinguish numeric prediction from classification; Calculate signed residuals with a stated convention. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Why does mean residual zero not imply perfect predictions?

Teacher check: Teacher explanation: Negative and positive residuals cancel even when individual errors are nonzero.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Regression and residuals. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Residuals -1, 1, 0 cm; the first record is overpredicted.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

MAE: original $5/3$ cm and changed $4/3$ cm. Signed mean reports average direction and can hide offsetting errors. Changed-case check: Residuals become $[-3, 0, 1]$; mean $= -2/3$ cm and MAE $= 4/3$ cm. Each residual decreases by 1 under the observed-minus-predicted convention.

Changed-case reasoning: Residuals become $[-3, 0, 1]$; mean $= -2/3$ cm and MAE $= 4/3$ cm. Each residual decreases by 1 under the observed-minus-predicted convention.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Residual Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Positive and negative residuals cancel, so their average proves accuracy.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Residuals -1, 1, 0 cm; the first record is overpredicted.

A stronger claim makes a stronger project: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: Residuals [-2, 1, 2] cm; mean $\frac{1}{3}$ cm. First overpredicted, others underpredicted.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

+3 cm means underprediction by 3 cm. State this sign convention consistently.

Independent question

Define residual as actual minus predicted. Actual height is 14 cm and prediction is 17 cm. Calculate residual and interpret its sign.

Expected independent answer

Residual is $14 - 17 = -3$ cm. The model overpredicts by 3 cm under the stated convention.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A negative residual means a negative physical height.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Residual=-3 cm means overprediction by 3 cm. Absolute error=3 cm. State the sign convention because some sources reverse it; keep units and convention consistent across plots and calculations.

Explanations for the additional workbook examples

W1 Calculate residuals

Compute the worked residuals and identify overprediction.

Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

W2 Explain cancellation

Why does mean residual zero not imply perfect predictions?

Teacher explanation: Negative and positive residuals cancel even when individual errors are nonzero.

W3 Read a pattern

Interpret residuals -2, 0, 3 ordered by growth stage.

Teacher explanation: They shift from overprediction to underprediction; investigate a missed relationship without claiming a proven cause.

W4 Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give residuals, mean residual and error directions.

Teacher explanation: Residuals [-2, 1, 2] cm; mean $1/3$ cm. First overpredicted, others underpredicted.

W5 Check units

A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

Teacher explanation: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

Assess independent transfer

Selected case: Observed [6, 10, 13], predicted [8, 9, 11] cm. Give residuals, mean residual and error directions.

Selected case answer

Teacher explanation: Residuals [-2, 1, 2] cm; mean $1/3$ cm. First overpredicted, others underpredicted.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Use a miniature growth-prediction side experiment to distinguish regression evidence from the main classifier metrics.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 20: Mean, Median, and Baselines. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions

without supplying the W 4 conclusion. Use regression and residual in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Residuals [-2, 1, 2] cm; mean 1/3 cm. First overpredicted, others underpredicted. Fresh retry observed [3, 8, 12], predicted [4, 8, 10]: residuals [-1, 0, 2], mean 1/3 in the target unit. Require sign interpretations. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Constant prediction baselines. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/20, ai-4/22, ai-4/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.